MIL-S-0010762D
24 March 1965
USED IN LIEU OF
MIL-S-10762C
13 November 1961

MILITARY SPECIFICATION

SEMITRAILER, REFRIGERATOR:
7 1/2-TON, 2-WHEEL, M349A1, M349A2, AND M349A3

This limited coordination military specification has been prepared by the U.S. Army Mobility Command (Army Tank-Automotive Center) based upon current available technical information, but it has not been approved for promulgation as a revision of Military Specification MIL-S-10762C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers three models of a 7 1/2 ton, 2-wheel, semitrailer with an integral, removable, self-contained refrigerating unit.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-C-520	-	Coating, Underbody (For Motor Vehicles).
VV-H-910	-	Hydraulic Fluid, Non-Petroleum Base, Automotive.
ZZ-I-550	-	Inner Tube, Pneumatic Tire.

Military

MIL-P-514	-	Plates; Identification, Transportation-Data, and Blank.
MIL-L-3150	-	Lubricating Oil, Preservative, Medium.
MIL-G-3278	-	Grease; Aircraft and Instruments (For Low and High Temperatures).
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-S-10379	-	Suppression, Radio Interference General Requirements for Vehicles (and Vehicular Sub-Assemblies).

FSC-2330

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MIL-R-10735 - Refrigeration Unit, Mechanical, Panel Type, Gasoline Engine Driven for Semitrailer Refrigerator.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-T-12459 - Tires, Pneumatic: For Military Ground Vehicles.
MIL-H-13910 - Hydraulic Fluid, Non-Petroleum Base, Automotive Brake, Arctic.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210 - Climatic Extremes for Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies: Preservation and Packaging of.
MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

ENGINEERING PARTS LIST
EPL 8736005 - Semitrailer, Refrigerator: 7 1/2 ton, 2-wheel, M349A1.
EPL 8736561 - Semitrailer, Refrigerator: 7 1/2 ton, 2-wheel, M349A2.
EPL 8736687 - Semitrailer, Refrigerator: 7 1/2 ton, 2-wheel, M349A3.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), a preproduction model(s) of the complete vehicle to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 Materials. Materials specified herein and in the referenced specifications, and standards (see 2.1), shall be free from all defects and imperfections that might affect the serviceability and function of the finished product.

3.2.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL's cited in 2.1, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as

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listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3 First production vehicle. The first production vehicle of each model to be delivered under the contract shall be examined by Government personnel to determine compliance with the requirements of this specification and for those examinations and tests specified herein. Vehicles submitted by the contractor shall be production models which are representative of the units to be delivered under the contract.

3.3.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein specified. Vehicles manufactured in accordance with this design shall be subjected to examinations and tests by the Government to prove vehicle capabilities and performance in accordance with the following:

(a) Vehicle shall require only organizational maintenance (first and second echelon) during 10,000 miles of normal operation.
(b) Vehicle shall require no replacemnt or major overhaul or any major automotive component during 20,000 miles of normal operation.

3.3.1.1 Reliability operations. To assure that workmanship and reliability requirements are met during the 20,000 miles of operation, the vehicle with a rated payload shall be capable of operating as follows:

(a) 19 percent on paved roads either concrete or asphalt or any combination of the two.
(b) 40 percent on level cross-country terrain.
(c) 40 percent on hilly cross-country terrain.
(d) 1 percent on belgian block.

3.4 Construction. Vehicles shall be constructed in accordance with the drawings, specifications, and standards listed or referred to in the Engineering Parts List (EPL's) specified in 2.1, and as supplemented herein. If more than 1 unit of a particular component is used in or on the vehicle, the components used shall be identical in make, material, and quality. The riveting, welding practices, and quality shall be the same on each vehicle.

3.4.1 Spare wheel assembly. One spare wheel, tire and tube assembly shall be furnished with each chassis.

3.4.2 Landing gear installation. The landing gear shall support the fully loaded semitrailer and withstand without damage, the strains imposed when coupling and uncoupling the semitrailer from the towing vehicle.

3.4.3 Tires. Tires shall conform to class MS, style a, of Specification MIL-T-12459.

3.4.4 Inner tubes. Inner tubes shall conform to Specification ZZ-I-550.

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3.4.5 Radio interference suppression. Vehicles shall be suppressed for radio interference in accordance with Specification MIL-S-10379 for tactical vehicles. Resistance type ignition cable, in lieu of suppressed spark plugs, is not acceptable.

3.4.6 Electrical. Exterior lighting system shall be 24-volts, interior lighting system shall be 12-volts, and shall conform to applicable drawings.

3.4.7 Refrigeration unit. The refrigeration unit, gasoline engine driven, conforming to Specification MIL-R-10735 will be furnished by the Government, and shall be installed by the contractor.

3.4.8 Wood treatment. All wood parts for vehicles furnished in accordance with this specification shall be cleaned and treated with wood preservative prior to painting. The wood surface shall be dry and thoroughly cleaned of dirt, grease, and other foreign matter before treatment. Where practicable, the wood parts shall be cut to final dimensions, and boring and shaping operations shall be completed prior to treatment.

3.5 Performance characteristics. Each semitrailer shall demonstrate vehicle performance characteristics, specified herein; when equipped as depicted on the applicable drawings, serviced with applicable products listed in table I, subjected to loading and operating conditions specified herein, and towed by:

(a) Truck-tractor - 2 1/2 ton, 6x6, M275
(b) Truck-tractor - 5 ton, 6x6, M52

3.5.1 Environmental. The vehicle shall operate in ambient air temperature of minus 40° to plus 125°F. The complete vehicle when in storage shall withstand climatic extremes as specified in Standard MIL-STD-210, without deterioration that may cause failure of any component part of the vehicle.

Table I. Service product specifications

Product use	Ambient air temperature operating range	
	-40° to +125°F.	-40° to -65°F.
Fluid:		
For hydraulic brakes	VV-H-910	MIL-H-13910
For shock absorbers	MIL-H-5606	MIL-H-5606
Grease:		
For sealed bearings	MIL-G-3278	MIL-G-3278
For general chassis lubrication including wheel bearings	MIL-G-10924	MIL-G-10924
Oil:		
For general purpose lubrication	MIL-L-3150	MIL-L-3150

3.5.2 Payload, towing speeds, and trailing ability.

3.5.2.1 Payload. The rated payload capacity shall be 15,000 pounds.

3.5.2.2 Gross weight. The gross weight of the fully equipped vehicle, including 15,000 pounds payload, shall not exceed 24,000 pounds.

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3.5.2.2.1 Weight distribution. The weight distribution shall, to the maximum practicable extent, insure uniform tire loading and shall not exceed 14,000 pounds at the axle.

3.5.2.2.2 Fifth wheel. The maximum load at the fifth wheel shall not exceed 10,000 pounds, when vehicle is loaded with rated payload.

3.5.2.3 Tracking. When concurrently transporting the maximum rated payload (see table II), and being towed over level, smooth, improved and prepared hard-surfaced roads at the appropriate sustained speed specified in table II, the trailer shall track the towing vehicle without weaving from side to side of the path of the towing vehicle in a straight line and without swaying laterally to the extent affecting the controllability of the vehicle combination.

3.5.2.4 Cross-country operation. When simultaneously carrying the maximum rated payload specified in table II, and being towed over unimproved roads, trails, open fields, rolling hills, and other cross-country terrain at the appropriate speed specified in table II, the semitrailer shall follow the towing vehicle without damage to the semitrailer or to the towing vehicle.

Table II. Loads and speeds

Maximum payload (pounds)	Highway	Maximum speed (mph)	
		Cross-country	Belgian block
15,000	50	25	15

3.5.3 Gradeability and stability.

3.5.3.1 Longitudinal grades. The semitrailer, loaded with cross-country payload at rated cross-country speed (see table II) and ascending or descending longitudinal grades of 60 percent, shall follow the prime mover without sliding or shifting to the extent that would cause losing control of vehicle combination.

3.5.3.2 Side slopes. The semitrailer shall follow the prime mover without slipping or upsetting when operating, either left or right, on side slopes having a 20 percent grade.

3.5.4 Turning ability. When coupled to the towing vehicle operating in its minimum turning circle, the semitrailer shall follow without cramping or side slipping. The semitrailer shall assume angle of 90 degrees to the towing vehicle without damage to either or interference between itself and the towing vehicle.

3.5.5 Braking ability.

3.5.5.1 Service brakes. The service brakes of the semitrailer and towing vehicle combination shall control, decelerate and stop the semitrailer within 30 feet from a speed of 20 miles per hour (mph) on dry, hard, level smooth surface roads.

3.5.5.2 Automatic actuation. The semitrailer shall be equipped with an automatically actuated device to apply the semitrailer brakes upon breakaway from the towing vehicle, and to maintain application of brakes for not less than fifteen minutes.

3.5.6 Deep fording. The semitrailer shall ford hard-bottomed, salt or fresh water crossings of sufficient depth to cover the top of the sub-frame members, and shall withstand such immersion for periods up to 15 minutes duration without damage to the semitrailer or its components.

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3.5.7 Wheel assembly balancing. Each assembly of tire, tube, wheel, and brake drum shall be balanced within practicable limits, so as to prevent wheel tramp or vibration at road speeds up to 50 mph.

3.5.8 Heat leakage. When tested as specified in 4.4.1.11, vehicle shall lose not more than 63 British thermal units per hour per degree Fahrenheit (°F.), at a temperature differential of not less than 5°F.

3.5.9 Body waterproofness. The semitrailer body and all components shall be waterproof to preclude the entrance of water due to heavy rains, melting snow, and road splash during inclement weather as well as to prevent the penetration of moisture resulting from other natural causes.

3.6 Treatment and painting. Treatment and painting shall be in accordance with Standard MIL-STD-193. Exterior color shall be olive drab with white markings.

3.7 Marking. Registration and other markings shall be in accordance with Standard MIL-STD-642. Loading limit and height lines shall be in accordance with the applicable drawing notes.

3.8 Name, shipping, and service data plates. Data plates shall be photographic aluminum (sensitized). The plates shall have the photosensitive silver compounds imbedded within the oxide layer; and after photographic processing, the letters, numbers and all other characters shall be sealed in the oxide layer by chemical treatment of the layer. The plates shall have a black background, except that danger and warning plates shall have a red background. The size of the plate, type and size of lettering, and the information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. Thickness of the plates shall be 0.030 inch minimum. The plates shall meet the performance requirements of Specification MIL-P-514.

3.9 Undercoating. Vehicles shall be undercoated using commercial sandless material meeting the requirements of Specification TT-C-520 for sag, acid and alkali resistance, corrosion, fire resistance, resistance to flow at elevated temperatures, salt-spray (fog) resistance, cold temperature adhesion, and abrasion resistance. Coating shall be applied to a thickness of 1/16 to 1/8 inch; to underbody and under chassis sheet metal surfaces having a thickness of 1/8 inch or less; except that drain holes, lubrication points, axle, brakes, and suspension components shall be kept free of coating material.

3.10 Service and adjusting. Prior to acceptance of the semitrailers by the Government inspector, contractor shall adjust and service each semitrailer for immediate operational use including at least the following: adjust brake system, check electrical system, inflate all tires, and completely lubricate chassis and all running gear with grades of lubricants recommended for the ambient air temperatures at the delivery point.

3.11 Workmanship. Workmanship shall be of a quality to assure delivery of semitrailers which are free from defects resulting from defective material, improper manufacturing or assembling practices, and which shall meet or exceed requirements specified herein. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent limitation is included in the approved quality control procedure. The contractor shall notify the Government and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government production inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Qualified products. The contractor's inspection records shall be checked to determine contractors conformance to 3.2.1. The contractor shall provide the Government with evidence that materials (see 3.2) used are as specified in the applicable specifications and drawings.

4.2 First production inspection.

4.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during fabrication of the first production vehicle to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.1.1.1 In-process inspection. An in-process inspection will be made at the contractor's or sub-contractor's plant, prior to the application of primer or paint.

4.2.1.2 Complete vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsequent to contractor's inspection at the contractor's plant to determine conformance to contract and specification requirements.

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4.2.1.2.1 Corrections. Corrections found necessary as a result of above inspection shall be made by the contractor on the vehicle.

4.2.1.3 Vehicle disposition. Upon completion of inspection and acceptance of the first production vehicle of the following vehicle models: M349A1, M349A2 and M349A3.

The vehicles shall remain at the manufacturing facility as a production sample. These vehicles may be released for shipment after 120 days from start of production has elapsed or when the production run of the particular model has been completed prior to the 120 day time limitation. If the quality level of the subsequent produced vehicles exceeds the acceptance standard, the first production vehicle may be released sooner at the discretion of the Contracting Officer. The contractor shall service and maintain the vehicles during this period in accordance with the applicable documents for care and preservation while in storage.

4.2.1.4 Production vehicle test. After inspection, correction and acceptance of the first production vehicle, two production vehicles of a model, as produced, shall be selected by the Government and shall be subjected to examinations and tests for conformance to this specification, table III - 20,000 mile test. The test will require a minimum of 60 days at a Government test site. The contractor shall support the test by providing any vehicle spare parts necessary for the conditions of completion of such a test. Delay in test completion caused by the vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall not be the basis for adjustment of the contract delivery schedule or contract price.

Table III. 20,000 mile test

Course	Mileage & Speeds	Vehicle loading
Hard Surface	4,000 miles at varying speeds up to 50 mph	15,000 lbs.
Level Cross Country	7,900 miles at varying speeds up to 25 mph	15,000 lbs.
Hilly Cross Country	7,900 miles at speeds applicable to the conditions of terrain	15,000 lbs.
Belgian Block	200 miles at varying speeds up to 15 mph	15,000 lbs.

4.2.1.4.1 Test failure. Failure of the vehicle to comply with any of the requirements specified or any deficiency of a workmanship or materials nature found on the vehicle during or as a result of the 20,000 mile test, will be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 20,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 20,000 mile test are similarly deficient, unless evidence satisfactory to the contracting officer is furnished by the contract that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

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4.3 Inspection provisions.

4.3.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production, submitted at one time for acceptance inspection.

4.3.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of Standard MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.3) to allow normal sampling in accordance with Standard MIL-STD-105.

4.3.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the applicable quality levels shown in table IV and table V. The following constitutes a part of the classification of defects (see 6.4).

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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Table IV. Major defects AQL = 25 defects/100 units

Def. No.	Characteristics	Major defects	Method of inspection
101.	Frame	Structural, welding, riveting, or sheet metal defects	Visual
102.	Fifth wheel plate and king pin	Improper mounting and assembly; welding defects	Visual
103.	Landing gear	Malfunction; extended, retracted, stowed or locking action	Visual and functional
104.	Axles	Improper assembly and mounting; welding defects; unusual noise	Visual and functional
105.	Service brakes	Malfunction; unusual noise, leaks; non-conformance to stopping distance	Visual and functional
106.	Automatic brake actuation	Malfunction; nonconformance	Functional
107.	Electrical system components and circuits	Malfunction, inoperative, damage	Visual and functional
108.	Suspension system components	Damage; leaks (see 6.4c)	Visual
109.	Wheels and tires	Tire damage	Visual
110.	Training ability	Nonconformance	Visual and functional
111.	Turning ability	Nonconformance	Visual and functional
112.	Overheating units		Tactile (touch)
113.	Batteries	Damage	Visual
114.	Towing speeds, highway	Nonconformance	Functional
115.	Refrigeration unit	Unusual noise; leaks (see 6.4a)	Visual and functional
116.	Fuel system components	Malfunction; damage; leaks (see 6.4a)	Visual and functional
117.	Engine	Malfunction; unusual noise; leaks (see 6.4b)	Visual and functional

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Table V. Minor AQL = defects/100 units

Def. No.	Characteristics	Minor defects	Method of inspection
201.	Wheels and tires	Improper size, type and mounting	Visual
202.	Landing gear	Improper assembly or installation	Visual
203.	Wiring or tubing	Improperly mounted; coded; or protective; defective	Visual
204.	Brake system components	Improper components assembly or installation, clearance inadequate	Visual
205.	Electrical system components	Improper installation or assembly	Visual
206.	Batteries and cables	Low electrolyte, or specific gravity; improper installations and cable lengths	Visual and hydrometer
207.	Bumper, lifting eyes and tiedowns	Missing; improperly located; welding defects	Visual and functional
208.	Body, doors, storage boxes	Improper fits, assembly, installation; defective materials	Visual and functional
209.	Refrigeration unit	Improper installation	Functional
210.	Lubricating fitting	Defective; missing; improper installation	Visual
211.	Paint	Application and color improper	Visual
212.	Decal markings, data and instruction plates	Incomplete data; missing; improper location or size	Visual
213.	Protective coatings	Application, coverage or materials improper	Visual
214.	Lubrication	Improper	Visual

4.3.2.1 Unclassified efects. All defects having no bearing on function, safety, interchangeability or lift, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day-period will be added to the minor classification with no increase to AQL's.

4.3.2.2 Recurring major deficiencies. A major deficiency (see 6.5) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.3.2.3 Recurring minor deficiencies. A minor deficiency (see 6.5) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiency. Correction of all deficiencies shall be accomplished prior to acceptance.

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4.4 Classification of tests.

(a) Acceptance tests, 4.4.1 through 4.4.1.12
(b) Control tests, 4.4.2 through 4.4.2.2
(c) Inspection comparison test (ICT), 4.4.3 through 4.4.3.6

4.4.1 Acceptance tests. After a 2 mile break-in, each vehicle shall be operated for a distance of not less than 5 miles by the contractor at the place of manufacture. Vehicles shall be serviced with specified products listed in table I. Vehicle(s) (less payload) shall pass the performance requirements specified herein. Each vehicle shall be driven in reverse gear a minimum of 50 feet. On completion of the 5 mile road test, doors shall be opened and closed to assure proper operation, fit, alinement and action of the control devices. In lieu of conducting the actual road test the contractor may determine conformance to the specification by simulating the road test conditions through the use of a six wheel drive chassis dynamometer. The contractor shall develop a detailed test procedure for the chassis dynamometer, and submit same to the procurement activity for review and approval. The procedure shall include the method that will be utilized to check vehicle capability of meeting tests specified in table VI.

4.4.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that the corrective action has been taken.

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Table VI. Location for tests

Title of tests	Test paragraph	Place of manufacture	Government proving ground
Production vehicle test	4.2.1.4		X
Test failure	4.2.1.4.1		X
Acceptance tests	4.4.1	X	
Test failure	4.4.1.1	X	
Construction check	4.4.1.2	X	X
Highway operation and tracking test	4.4.1.3	X	X
Turning test	4.4.1.4	X	X
Braking test	4.4.1.5	X	X
Automatic actuation device check	4.4.1.6	X	X
Landing gear test	4.4.1.7	X	X
Wood treatment check	4.4.1.8	X	X
Wheel balancing check	4.4.1.9	X	X
Body waterproofness test	4.4.1.10.1	X	X
Floor waterproofness test	4.4.1.10.2	X	X
Test failure	4.4.1.10.3	X	X
Heat leakage test	4.4.1.11	X	X
Painting and marking check	4.4.1.12	X	X
Control test	4.4.2	X	
Frequency	4.4.2.1	X	
Failure	4.4.2.2	X	
Inspection comparison tests	4.4.3		X
Test failure	4.4.3.1		X
Environmental test	4.4.3.2		X
Cross-country operational test	4.4.3.3		X
Longitudinal grade and side slope operational test	4.4.3.4		X
Fording tests	4.4.3.5		X
Radio interference suppression test	4.4.3.6		X

4.4.1.2 Construction check. To determine conformance to 3.4, during production of vehicle, it shall be checked for assembly.

4.4.1.3 Highway operational and tracking test. To determine conformance to 3.5.2.3, the vehicle shall be attached to prime mover and towed as specified and checked for trailing in a straight line.

4.4.1.4 Turning test. To determine conformance to 3.5.4, the vehicle combination shall be turned to the right and left while being operated in forward and reverse to confirm the performance requirements.

4.4.1.5 Braking tests. To determine conformance to 3.5.5.1, the vehicle combination shall be operated as specified and stopping performance requirements checked.

4.4.1.6 Automatic actuation device check. To determine conformance to 3.5.5.2, the vehicle combination shall be operated and the automatic actuation device actuated and checked for functional requirements.

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4.4.1.7 Landing gear test. To determine conformance to 3.4.2, the vehicle shall be placed on firm surface and the landing gear (landing legs and leveling jacks) of each vehicle shall be operated and checked for functional requirements.

4.4.1.8 Wood treatment check. To determine conformance to 3.4.8 through 3.3.8.4, all wood parts shall be checked for preparation and painting, for proper application.

4.4.1.9 Wheel balancing check. To determine conformance to 3.5.7, the wheels shall be checked for balance.

4.4.1.10 Waterproofness tests.

4.4.1.10.1 Body waterproofness test. Each vehicle van body shall be subjected to a fifteen minute spray test to determine waterproofness of panel joints and door seals. Test shall be conducted prior to installation of inner panels or insulation of the van body. The spray shall be delivered by nozzles operating at (25) pounds per square inch (psi) water pressure, sufficient in number and so placed (approximately 2 feet from van body) to afford full coverage of sides, roof, front and rear of van body. During test, doors shall be closed as in a normal operation.

4.4.1.10.2 Floor waterproofness test. The floor shall be tested for water tightness and evidence of water leakage shall be cause for rejection until permanent repairs are made and no leakage exists.

4.4.1.10.3 Test failure. If leaks occur that cannot be corrected by simple adjustment, the van body shall be rejected. Upon completion of repairs the van body shall be retested as specified.

4.4.1.11 Heat leakage test. The production vehicles 4.2.1.4, control test vehicles 4.4.2 and inspection comparison test vehicles 4.4.3, shall be subjected to the heat leakage test. Vehicles shall be equipped with a refrigerating unit and placed in a controlled temperature room, where the temperature difference between inside and outside of vehicle shall be a minimum of 50°F. during any test. Ten thermocouples or thermometers which have been checked against a secondary standard thermometer shall be placed outside and not less than 5 inches from body of the vehicle and located as follows to record the ambient air temperature: one at each of the eight corners; one in the center of the top; and one at the center of the bottom. Thermometers shall be not less than 5-inches from body of vehicle. Electric resistors surrounded by shields to minimize radiation shall be placed within vehicle. If incandescent heating elements are used, they shall be operated at 1/2-normal voltage by connection two in series. Ten thermocouples which have been checked against a secondary thermometer shall be placed inside vehicle as follows: One in each of two opposite upper and two opposite lower corners; one in the center of each side; one in the center of the ceiling; one in the center of the front wall; one in the center of the door; one on the center line of the floor 4 feet from the door; and one on the center of the floor 3 feet from the front wall. Each inside thermocouple shall be placed to measure the air temperature 1 inch from the surface to which it is attached. An 8- to 10-inch electric fan shall be placed within the semitrailer on the raised portion of the floor and faced toward the door to provide interior air circulation. The door shall then be closed and not opened for the duration of the test. Electric current shall then be sent, through the resistor and the fan to

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provide 900 to 1,000 watts within the semitrailer. When the interior temperature has reached a stationary level, the amount of heat required to hold this level for 6 hours in the fixed ambient air temperature shall be measured in terms of watt hours. The heat leakage of the trailer shall then be determined as follows:

$$\text{Heat Leakage} = \frac{\text{Watt hours} \times 3.415}{6 \times (\text{interior temperature} - \text{exterior temperature})}$$

4.4.1.12 Painting and marking check. To determine conformance to 3.6 and 3.7, all vehicles shall be examined for proper painting and marking.

4.4.2 Control tests. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.4.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles, produced, and thereafter one per month, shall be checked in accordance with 4.4.1 through 4.4.1.12.

4.4.2.2 Failure. If the vehicle selected fails to pass any of the control tests the Government inspector shall stop acceptance examination and testing on subsequent lots until such time as conditions causing the failure have been remedied.

4.4.2.3 Sixty mile test. Each vehicle selected in accordance with 4.4.2.1 shall be loaded with highway payload or with a simulated load of equal weight and driven for a distance of 60 miles and subjected to tests specified therein.

4.4.3 Inspection comparison tests. The Government may select vehicles any time during the contract production period and subject vehicle to tests listed in table VI and table VII to reveal deficiencies of manufacturer or design that may reduce the effective operation of these items in the field and to compare existing quality with previous atandards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be loaded with loads specified in table II. Vehicle selected shall not include any vehicles previously tested for conformance to 4.4.2(Control tests).

4.4.3.1 Test failure. Failure of any vehicle tested as specified herein to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

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Table VII. 4150 miles

Course	Mileage & Speeds	Vehicle payload
Paved surface	900 miles; up to 50 mph	15,000 lbs.
Level Cross-Country	1,500 miles, as applicable to conditions of terrain, up to 25 mph	15,000 lbs.
Hilly Corss-Country	1,500 miles, as applicable to conditions of terrain	15,000 lbs.
Belgian Block	150 miles, as applicable to conditions of terrain 70 to 15 mph	15,000 lbs.

4.4.3.2 Environmental test. To determine conformance to 3.5.1, the vehicle selected for this test shall be tested under applicable climatic conditions or as specified by procuring activity to assure vehicle meets performance requirements.

4.4.3.3 Cross-country operational test. To determine conformance to 3.5.2.4, each vehicle selected for this test shall be loaded with rated cross-country payload or a simulated load of equal weight and driven as specified and checked for performance requirements.

4.4.3.4 Longitudinal grade and side slope operational test. To determine conformance to 3.5.3.1 and 3.5.3.2, each vehicle selected for this test shall be loaded with rated cross-country payload or a simulated load of equal weight and driven on specified grades and slopes and checked for performance requirements.

4.4.3.5 Fording tests. To determine conformance to 3.5.6, the vehicle shall be towed through water crossings at depth specified, and shall be immersed in water for the required time, to prove fording ability without subsequent damage to vehicle.

4.4.3.6 Radio interference suppression test. To determine conformance to 3.4.5 the vehicle shall be subjected to a radio interference suppression test.

4.4.4 Preservation, packaging and vehicle processing. Material and equipment shall be inspected prior to shipment to assure conformance to the applicable requirements specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Vehicles shall be preserved and packaged for shipment and storage in accordance with Standard MIL-STD-281, as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Vehicle covered by this specification, equipped with a removable, self-contained refrigerating unit, is intended for transporting military supplies and equipment. The vehicle is designed for use with prime movers:

(a) Truck-tractor 2 1/2 ton, 6x6, M275.
(b) Truck-tractor 5 ton, 6x6, M52.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Applicable requirements for treatment and painting of vehicles (see 3.6).
(d) Information to be shown on data plates (see 3.8).
(e) Level of preservation and packaging and vehicle preparation for shipment and storage data sheet, or other implementing instructions, to indicate requirements of Standard MIL-STD-281 that apply to the vehicle being procured.

6.3 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.4 The following definitions shall be used for the classification of defects for leaks:

(a) Weep: Any evidence of fluid beyond the seal.
(b) Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
(d) Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.5 Definitions of recurring major and minor defects. The Government inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The Government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle and specification and definitions contained in Standard MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodians:
Army - MO

Preparing activity:
Army - MO

Project No. 2330-A327

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